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# Progress on
# Sustainable Development Goals:
# Education & Youth Employment Indicators

**SDG 4:** Quality Education

**SDG 8:** Decent Work & Economic Growth

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# 1. Introduction

In this assignment, we have focused on creating a dashboard that visualizes the global progress of the selected United Nations Sustainable Development Goals. We have used real SDG data to make predictions in order to make real-life decisions. This process goes beyond visualizations because we convert complex and multi-indicator development data into clean and well-structured data by using all the concepts, we have learnt so far.

This analysis is based on Quality Education (SDG 4) and Decent work and Economic growth (SDG 8) and this helps to understand the correlation of both factors. Even though there are 17 SDGs in total, all those are divided into indicators, which makes a total of 169 indicators. All the raw data was extracted from the official UN SDG website's data portal. Then the datasets were cleaned and standardized using SQL Server Management Studio The created database was connected to Power BI and were modelled into a star schema. Advanced DAX Measures were used to create better visualizations that analyze important trends.

The final output was a five-page interactive dashboard with multiple visualizations that presents global insights across multiple perspectives. Buttons and filters were also added for accessibility. This dashboard can be used to analyze all the key trends throughout the years and make predictions.

# 2. Dataset Selection and Acquisition

## 2.1 Data Source: UN SDG Data Repository

All datasets were downloaded from the official United Nations SDG Data Platform (https://unstats.un.org/sdgs/dataportal), which is the primary source of tracking global progress for the 17 Sustainable Development Goals and 169 targets. This platform compiles data from National Statistical Offices, International Organizations, and specialized Agencies of the United Nations to ensure consistency with internationally agreed methodologies.

## 2.2 Indicator 4.1.1: Percentage of Children and Youth Reaching Minimum Proficiency Levels

**Official Definition**:

Percentage of children and young people who reached a minimum level of attainment in (a) reading and (b) mathematics by the end of the primary and lower secondary levels of education, by gender.

**Metadata:**

- **Goal**: SDG 4 - Quality Education
- **Target**: 4.1 - By 2030, ensure that all girls and boys are completely free, equitable and quality primary and secondary education leading to relevant and effective learning outcomes
- **Indicator Code**: 4.1.1
- **Custodian Agency**: UNESCO Institute for Statistics (UIS)
- **Frequency**: Annual (subject to national assessment cycles)
- **Unit of Measurement**: Percentage (0-100)
- **Disaggregation**: Sex, Education Level (Primary/Lower Secondary), Type of Skill (Reading/Mathematics)

**Coverage in This Analysis:**

- Nations: 182 (All countries with available data)
- Years: 2000-2023 (varying availability by country)
- Demographic Dimensions:
    - Sex: Both, Male, Female
    - Education Level: Primary (PRIMAR), Lower Secondary (LOWSEC
    - Skill Category: Reading, Mathematics

## 2.3 Indicator 8.6.1: Youth not in Education, Employment or Training (NEET) rates (%)

**Official Definition:** Percentage of young people aged 15-24 who are not in education, employment or training.

**Metadata:**

- **Goal**: SDG 8 - Decent Work and Economic Growth
- **Target**: 8.6 - By 2020, greatly lower the rate of youth not in jobs, school, or training.
- **Indicator Code**: 8.6.1
- **Custodian Agency**: International Labor Organization (ILO)
- **Frequency**: Annual (based on labor force surveys)
- **Unit of Measurement**: Percentage (0-100)
- **Disaggregation**: Sex, Age Group (15-24, 25-34, 35-44)

**Coverage in This Analysis:**

- Nations: 182 (overlap with SDG 4 data set)
- Years: 2000-2023
- Demographic Dimensions:
    - Sex: Female, Male, Both
    - Target Age Groups: 15-24 (primarily), 25-34, 35-44

## 2.4 Indicator Selection

Both indicators were chosen for several strategic reasons:

- Complementarity: Education and employment can be viewed as a process of stages in human capital development
- Policy Relevance: Both are important for development agencies and governments
- Data Quality: Comparatively broad coverage with standardized measurement approaches
- Equity Focus: Both are disaggregated by gender, allowing for the examination of existing differences
- 2030 Urgency: Both have "specific targets with looming deadlines that call for rapid implementation"

# 3. Data Preparation: From Raw Files to SQL Server Database

## 3.1 Initial Data Format and Conversion

The data was retrieved from the UN SDG Data Platform website in XLSX (Microsoft Excel) form, which is the official distribution format for the data repository. Data in each file had multiple columns representing geographic information, time series information, data values, as well as disaggregation variables.

**File Preparation Process:**

1. The XLSX Formatting: The XLSX files were then converted into CSV (Comma-Separated Values) format since it is easier to import the files as flat files.
2. Column Structure Validation: Before importing, the column names were checked to ensure consistency.

    - Goal
    - Indicator
    - SeriesCode
    - SeriesDescription
    - GeoAreaCode (Country Code)
    - GeoAreaName (Country Name)
    - TimePeriod (Year)
    - Value (Indicator Value)
    - Nature (Geographic Level: C=Country, R=Region, G=Global)
    - Sex
    - Additional disaggregation columns (varying by indicator)

3. Data Volume Assessment:

    - **RAW_Goal4**: Approximately 15,000+ rows covering education proficiency data
    - **RAW_Goal8**: Approximately 12,000+ rows covering NEET rates
    - **Combined**: ~27,000 raw records before cleaning

## 3.2 SQL Server database creation

A database named 'SDG_Analytics_DB' was created to store all data structures so that it ensures the separation of this assignment's data.

```
CREATE DATABASE SDG_Analytics_DB;
GO
USE SDG_Analytics_DB;
GO
```

## 3.3 Flat File Import Process

Two staging tables were created by importing the CSV files into SQL Server using SQL Server Management Studio (SSMS) Import Wizard.

- RAW_Goal4
- RAW_Goal8

# 4. SQL Server Data Architecture and Transformation

## 4.1 Staging Data: Unified Data Consolidation

A new table (Staging_SDG_Data) table was created to merge both Goal 4 and Goal 8 data.

```sql
-------------------------------------------------------------
-- STEP 1: CREATE THE UNIFIED STAGING TABLE
-------------------------------------------------------------
CREATE TABLE Staging_SDG_Data (
    FactID BIGINT PRIMARY KEY IDENTITY(1,1),
    Goal INT NOT NULL,
    Indicator NVARCHAR(50) NOT NULL,
    SeriesCode NVARCHAR(50) NOT NULL,
    SeriesDescription NVARCHAR(500),
    CountryCode NVARCHAR(3) NOT NULL,
    CountryName NVARCHAR(100) NOT NULL,
    DataYear INT NOT NULL,
    IndicatorValue FLOAT,
    Sex NVARCHAR(10),
    Education_Level NVARCHAR(50), -- Used Education_level
    Skill_Type NVARCHAR(50),      -- Used Type_of_skill
    Age_Group NVARCHAR(50),       -- Used Age
    Observation_Status NVARCHAR(10), -- Used Observation_Status
    Data_Source NVARCHAR(255)
);
GO
```

Key Design Decisions:

- NVARCHAR(MAX) for SeriesDescription and Data_Source to prevent truncation of lengthy text fields
- Single schema that fits all the disaggregation dimensions: Education_Level, Skill_Type, Age_Group, leaving those that do not apply as NULL.
- Identity column (FactID) preserved through to fact table for traceability

## 4.2 Data Loading with Initial Filtering

### Goal 4

```sql
INSERT INTO Staging_SDG_Data (
    Goal, Indicator, SeriesCode, SeriesDescription, CountryCode, CountryName, DataYear, IndicatorValue,
    Sex, Education_Level, Skill_Type, Age_Group, Observation_Status, Data_Source
)
SELECT
    CAST(Goal AS INT),
    CAST(Indicator AS NVARCHAR(100)),  -- match updated staging table
    SeriesCode,
    SeriesDescription,                  -- NVARCHAR(MAX) now, safe for long text
    CAST(GeoAreaCode AS NVARCHAR(3)),
    GeoAreaName,
    CAST(TimePeriod AS INT),
    CAST(Value AS FLOAT),
    Sex,
    Education_level,
    Type_of_skill,
    NULL,       -- Age_Group not applicable for Goal 4
    NULL,       -- Observation_Status not applicable for Goal 4
    Source
FROM
    RAW_Goal4
WHERE
    Nature = 'C'  -- Country-level data only
    AND Value IS NOT NULL;
GO
```

### Goal 8

```sql
INSERT INTO Staging_SDG_Data (
    Goal, Indicator, SeriesCode, SeriesDescription, CountryCode, CountryName, DataYear, IndicatorValue,
    Sex, Education_Level, Skill_Type, Age_Group, Observation_Status, Data_Source
)
SELECT
    CAST(Goal AS INT),
    CAST(Indicator AS NVARCHAR(100)),   -- match staging table
    SeriesCode,
    SeriesDescription,
    CAST(GeoAreaCode AS NVARCHAR(3)),
    GeoAreaName,
    CAST(TimePeriod AS INT),
    CAST(Value AS FLOAT),
    CASE
        WHEN Sex = 'BOTHSEX' THEN 'Total'
        WHEN Sex = 'FEMALE' THEN 'Female'
        WHEN Sex = 'MALE' THEN 'Male'
        ELSE 'Unknown'
    END,
    NULL,                               -- Education_Level not applicable for Goal 8
    NULL,                               -- Skill_Type not applicable for Goal 8
    Age,                                -- Age column from RAW_Goal8
    Observation_Status,                 -- A = Normal, U = Low reliability
    Source
FROM
    RAW_Goal8
WHERE
    Nature = 'C'                        -- Country-level data only
    AND Value IS NOT NULL;
GO
```

Filtering Criterion:

- Nature = 'C': includes only country level data, as regional and global aggregates are included to prevent double counting
- Value IS NOT NULL: Removes records without indication values
- Sex Standardization: Transforms Coded values (BOTHSEX) to readable labels at load

## 4.3 Comprehensive Data Cleaning

```sql
-- 1⃣ Standardize Sex values
UPDATE Staging_SDG_Data
SET Sex = CASE
    WHEN Sex = 'BOTHSEX' THEN 'Total'
    WHEN Sex = 'FEMALE' THEN 'Female'
    WHEN Sex = 'MALE' THEN 'Male'
    ELSE 'Unknown'
END;

-- 2⃣ Remove rows with low reliability
DELETE FROM Staging_SDG_Data
WHERE Observation_Status = 'U';

-- 3⃣ Optional: Standardize Age_Group codes (if needed)
-- Example mapping (uncomment and edit if you want to convert codes to ranges)
UPDATE Staging_SDG_Data
SET Age_Group = CASE
    WHEN Age_Group = 'N' THEN '15-24'
    WHEN Age_Group = 'A' THEN '25-34'
    WHEN Age_Group = 'G' THEN '35-44'
    ELSE Age_Group
END;

-- 4⃣ Verify final counts
SELECT Goal, COUNT(*) AS Rows_Per_Goal
FROM Staging_SDG_Data
GROUP BY Goal;
```

**Results After Cleaning:**

- Goal 4: ~13,000 rows
- Goal 8: ~10,500 rows
- Total: ~23,500 cleaned records

## 4.4 Star Schema Implementation

**Creating Dimension Tables**

```sql
-- Dim_Country
CREATE TABLE Dim_Country (
    CountryKey INT PRIMARY KEY IDENTITY(1,1),
    CountryCode NVARCHAR(3) NOT NULL UNIQUE,
    CountryName NVARCHAR(100) NOT NULL
);
INSERT INTO Dim_Country (CountryCode, CountryName)
SELECT DISTINCT CountryCode, CountryName FROM Staging_SDG_Data;
GO

-- Dim_Time
CREATE TABLE Dim_Time (
    TimeKey INT PRIMARY KEY,
    DataYear INT NOT NULL,
    Is_Baseline_Year BIT DEFAULT 0
);
INSERT INTO Dim_Time (TimeKey, DataYear, Is_Baseline_Year)
SELECT DISTINCT DataYear, DataYear,
    CASE WHEN DataYear = 2015 THEN 1 ELSE 0 END
FROM Staging_SDG_Data;
GO

-- Dim_Indicator
CREATE TABLE Dim_Indicator (
    IndicatorKey INT PRIMARY KEY IDENTITY(1,1),
    Goal INT NOT NULL,
    SeriesCode NVARCHAR(50) NOT NULL UNIQUE,
    IndicatorName NVARCHAR(500) NOT NULL
);
INSERT INTO Dim_Indicator (Goal, SeriesCode, IndicatorName)
SELECT DISTINCT Goal, SeriesCode, SeriesDescription
FROM Staging_SDG_Data;
GO
```

## Creating the Fact Table

```sql
CREATE TABLE Fact_SDG_Data (
    FactID BIGINT PRIMARY KEY,
    CountryKey INT FOREIGN KEY REFERENCES Dim_Country(CountryKey),
    IndicatorKey INT FOREIGN KEY REFERENCES Dim_Indicator(IndicatorKey),
    TimeKey INT FOREIGN KEY REFERENCES Dim_Time(TimeKey),
    Sex NVARCHAR(10),
    Education_Level NVARCHAR(50),
    Skill_Type NVARCHAR(50),
    Age_Group NVARCHAR(50),
    IndicatorValue FLOAT NOT NULL
);

INSERT INTO Fact_SDG_Data (FactID, CountryKey, IndicatorKey, TimeKey, Sex, Education_Level, Skill_Type, Age_Group, IndicatorValue)
SELECT
    s.FactID,
    dc.CountryKey,
    di.IndicatorKey,
    dt.TimeKey,
    s.Sex,
    s.Education_Level,
    s.Skill_Type,
    s.Age_Group,
    s.IndicatorValue
FROM Staging_SDG_Data s
JOIN Dim_Country dc ON s.CountryCode = dc.CountryCode
JOIN Dim_Indicator di ON s.SeriesCode = di.SeriesCode
JOIN Dim_Time dt ON s.DataYear = dt.DataYear;
GO
```

Designing Criterion:

- Degenerate Dimensions: Sex, Education_Level, Skill_Type and Age_Group maintained in fact table (low cardinality, often filtered in combination)
- Foreign Key Constraints: Enforce referential integrity
- Surrogate Keys: Tremendous in joins and future flexibility

## 4.5 Final Optimizations

**Additional Data Quality Check, Metadata Enhancement, Staging Cleanup and Final Verification**

```sql
-- Remove rows where IndicatorValue is NULL
DELETE FROM Fact_SDG_Data
WHERE IndicatorValue IS NULL;

-- Update UnitOfMeasure based on SeriesCode / Goal
UPDATE Dim_Indicator
SET UnitOfMeasure = CASE
    WHEN SeriesCode LIKE '4%' THEN 'Percent'
    WHEN SeriesCode LIKE '8%' THEN 'Percent'
    ELSE 'Unknown'
END;
```

```sql
DROP TABLE Staging_SDG_Data;
GO

-- Verification
SELECT 'Fact Table Rows' AS TableName, COUNT(*) AS CountValue FROM Fact_SDG_Data
UNION ALL
SELECT 'Dim Country Rows', COUNT(*) AS CountValue FROM Dim_Country
UNION ALL
SELECT 'Dim Indicator Rows', COUNT(*) AS CountValue FROM Dim_Indicator
UNION ALL
SELECT 'Dim Time Rows', COUNT(*) AS CountValue FROM Dim_Time;
GO
```

**Final Database Statistics:**

- Fact_SDG_Data: 23,487 rows
- Dim_Country: 182 rows
- Dim_Indicator: Multiple series (varying by disaggregation)
- Dim_Time: 24 rows

## 4.6 Summary of SQL Implementation

SQL development process successfully,

- Merged the two heterogeneous datasets into one unified structure.
- Applied different data quality filters.
- Standardized categorical values (sex, age groups) across indicators for comparison.
- Designed an optimal star schema with referential integrity.
- Prepared clean and structured data to create a dashboard using Power BI.

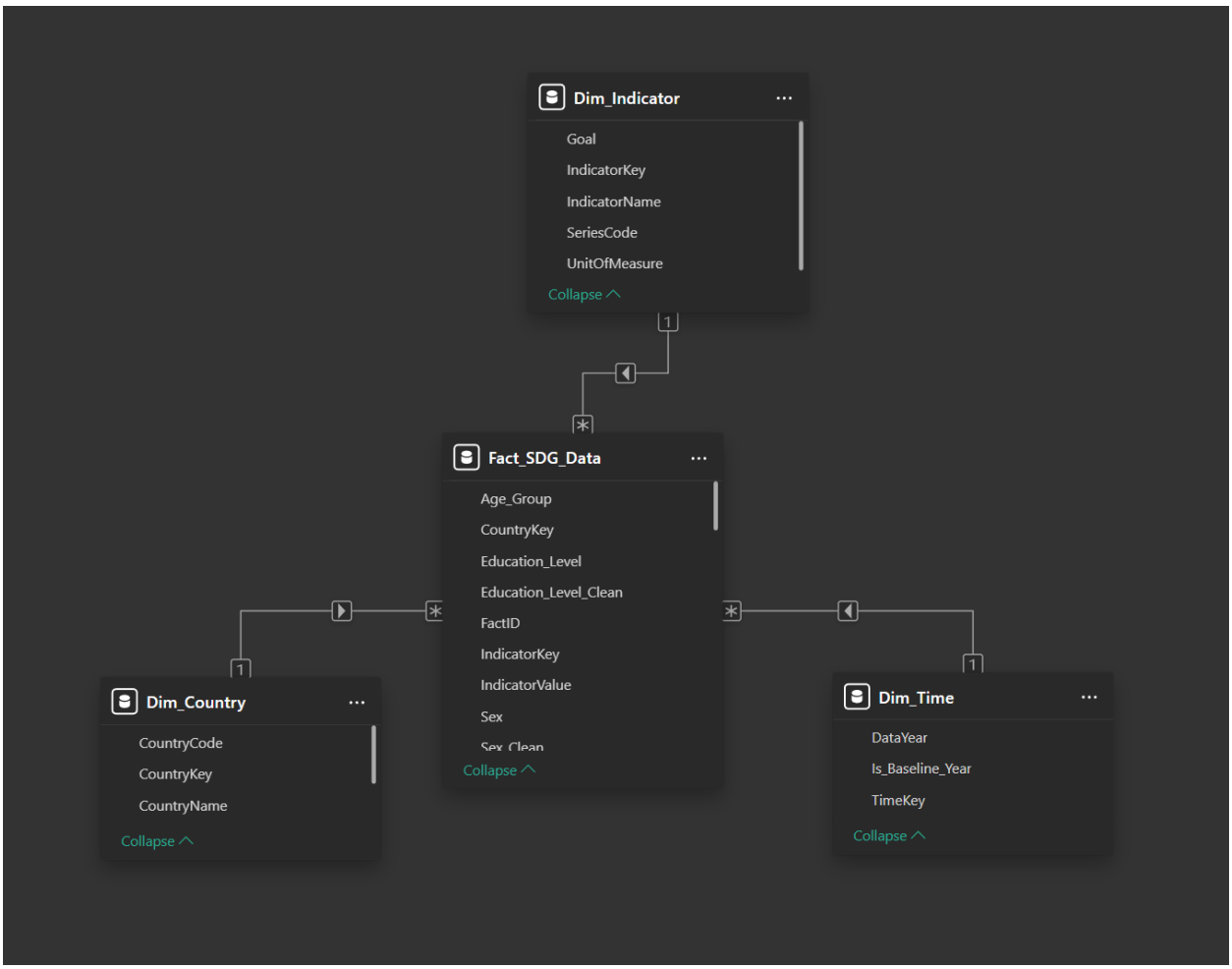
# 5. Power BI Data Modeling and Transformation

## 5.1 Database Connection and Import Strategy

The datasets were imported to PowerBI by connecting the SQL server to PowerBI Desktop. Fact_SDG_Data, Dim_Country, Dim_Time, Dim_Indicator were the tables (datasets) which were imported to PowerBI.

## 5.2 Data Model Relationships

Power BI automatically detected relationships based on the SQL Server foreign key constraints, creating a star schema data model:



## 5.3 Power Query Transformations

**Categorical Value Standardization**

Created calculated columns to ensure consistent, user-friendly labels:

```
Sex_Clean =
SWITCH(
    TRUE(),
    ISBLANK(Fact_SDG_Data[Sex]), "Not Specified",
    Fact_SDG_Data[Sex] = "Female", "Female",
    Fact_SDG_Data[Sex] = "Male", "Male",
    Fact_SDG_Data[Sex] = "Total", "Both",
    Fact_SDG_Data[Sex] = "Unknown", "Not Specified",
    Fact_SDG_Data[Sex]
)
```

```
Education_Level_Clean =
SWITCH(
    TRUE(),
    ISBLANK(Fact_SDG_Data[Education_Level]), "Not Specified",
    Fact_SDG_Data[Education_Level] = "LOWSEC", "Lower Secondary",
    Fact_SDG_Data[Education_Level] = "PRIMAR", "Primary",
    Fact_SDG_Data[Education_Level]
)

```

**Column Visibility Management and Data type Verification**

Calculated columns were used in all the visualizations to prevent confusions, and all the data types were changed accordingly in order to ensure proper aggregation and filtering.

## 5.4 Advanced DAX Measure Development

**Overview of Analytical Measures**

The sophisticated visualizations are made by using advanced DAX measures along with the other variables. This dashboard is created using 48 measures along with the 2 calculated columns and other variables. Those measures can be divided into 6 main categories.

- Basic Aggregations: Gender-specific averages and gaps
- Temporal Analysis: Baseline comparisons, year-over-year changes, total change calculations
- Performance Standardization: Normalized scores for cross-goal comparisons
- Predictive Metrics: 2030 projections and target achievement status
- Equity Metrics: Gender parity indices and gap calculations
- Ranking and Comparative: Country rankings and top performer identification

```
Education_Gap =
VAR HighestEd =
    CALCULATE(
        AVERAGE(Fact_SDG_Data[IndicatorValue]),
        Fact_SDG_Data[Education_Level] = "UPPSEC"  // Upper secondary
    )
VAR LowestEd =
    CALCULATE(
        AVERAGE(Fact_SDG_Data[IndicatorValue]),
        Fact_SDG_Data[Education_Level] = "PRIMAR"  // Primary
    )
RETURN
    HighestEd - LowestEd
```

```
Years_To_Target =
VAR CurrentValue = [Latest_Value]
VAR TargetValue = [Target_Value]
VAR Rate = [Annual_Change_Rate]
RETURN
IF(
    OR(
        ISBLANK(TargetValue),
        OR(ISBLANK(CurrentValue), ISBLANK(Rate))
    ),
    BLANK(),
    IF(
        Rate = 0,
        BLANK(),
        DIVIDE(TargetValue - CurrentValue, Rate)
    )
)

```

All of these measures transform raw data into actionable insights which enables the dashboard to answer critical questions.

# 6. Dashboard Architecture and Visualization Strategy

## 6.1 Page Structure and Navigation Design

This is a well-structured Power BI dashboard with 5 pages in total. It supports deep and progressive analytical data exploration by addressing a distinct analytical objective. It also maintains consistent filters and color schemes to ensure usability.

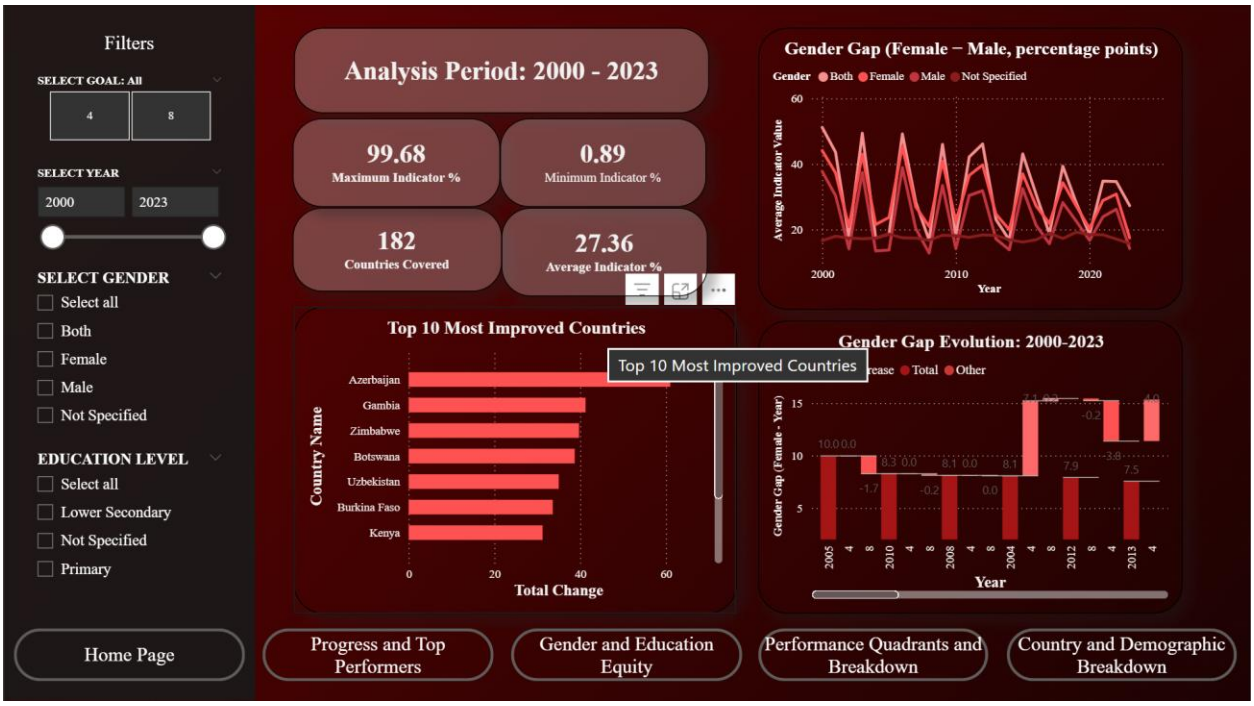
The dashboard consists of 5 main pages.

- Page 1 - Home page
- Page 2 - Progress and Top Performers
- Page 3 - Gender and Education Equity
- Page 4 - Performance Quadrants and Breakdown
- Page 5 - Country and Demographic Breakdown

### 6.1.1 Page 1 – Home Page

Serves as an introduction and provides buttons to navigate to the other pages.

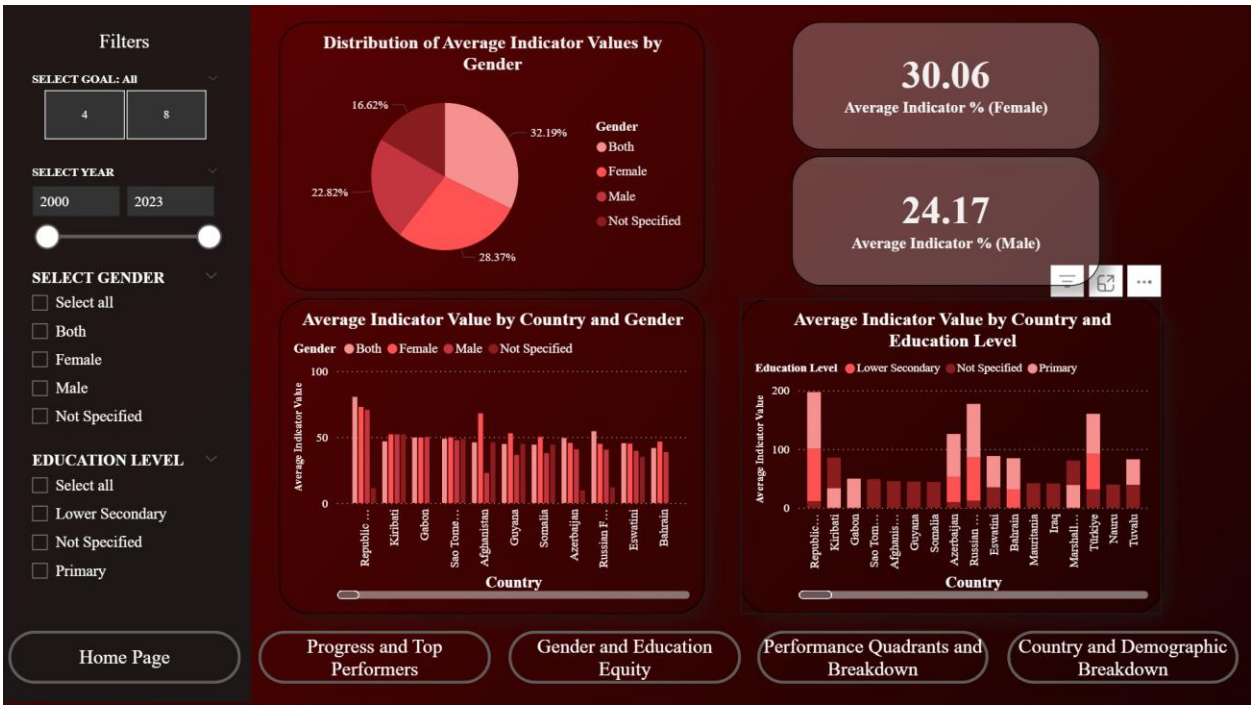
## 6.1.2 Page 2 - Progress and Top Performers




**Visuals and Graph Explanations:**

- **Summary KPI Cards:** Display overall country coverage and minimum/maximum indicator values. Designed for quick comprehension of dataset scope.
- **24-Year Trend Lines by Gender:** Line charts track educational attainment and youth employment over time, illustrating temporal trends and gender disparities.
- **Top 10 Most Improved Countries:** Bar chart ranking countries based on cumulative indicator improvements. Highlights best-practice examples for benchmarking.
- **Gender Gap Evolution (Waterfall Chart):** Visualizes cumulative changes in gender parity, making patterns of improvement or decline clear over time.

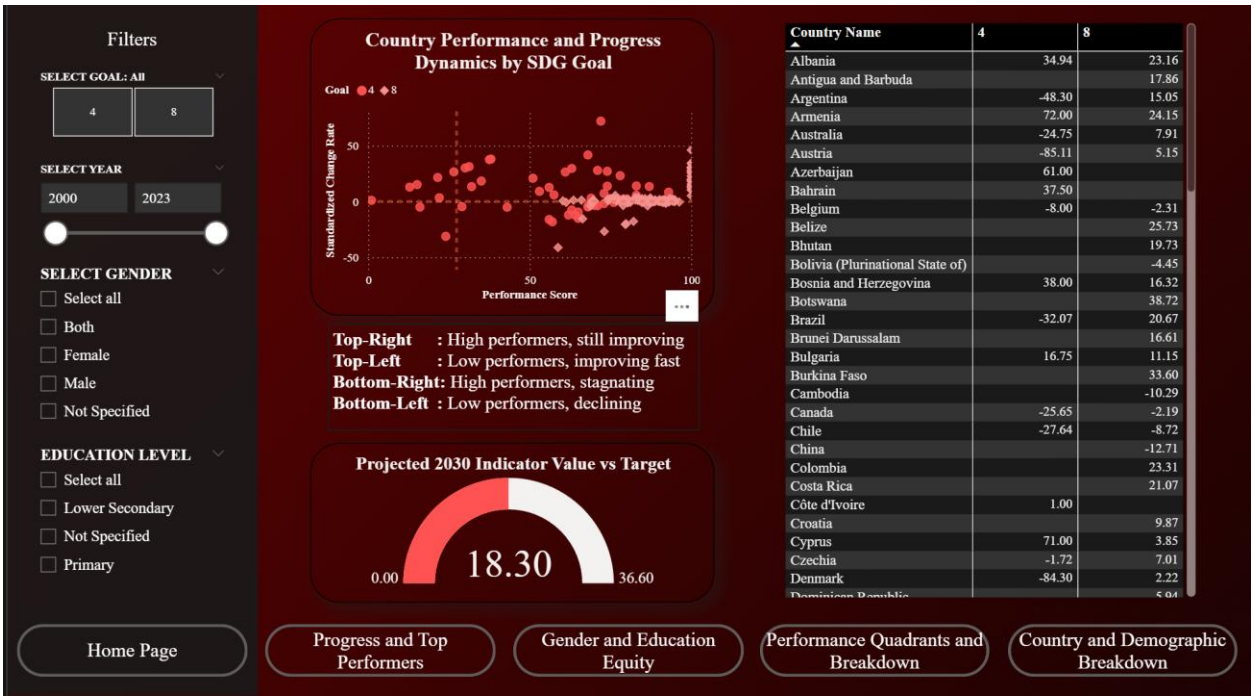
### 6.1.3 Page 3 - Gender and Education Equity




**Visuals and Graph Explanations:**

- **Distribution by Gender (Pie Chart):** Provides context for population or sample distribution.
- **Gender-Specific KPI Cards:** Show comparative performance between males and females across indicators.
- **Country Comparisons by Gender and Education Level:** Enables identification of specific disparities at the intersection of country and education stage.

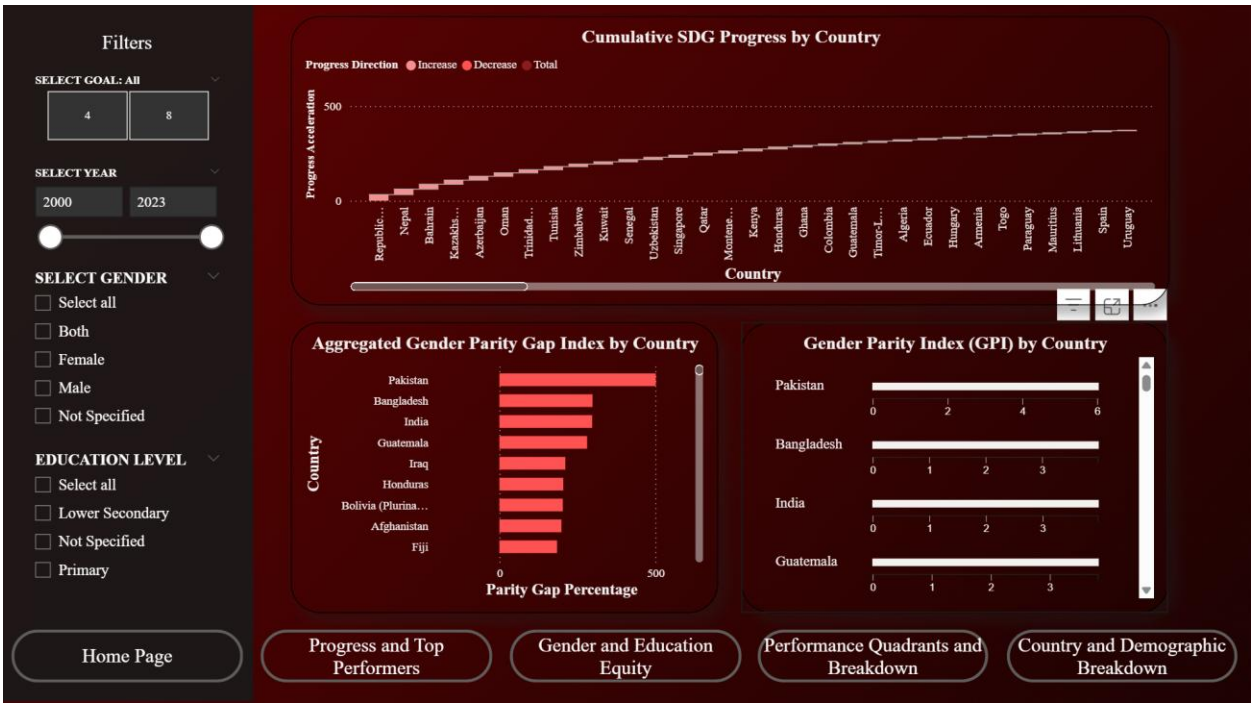
## 6.1.4 Page 4 - Performance Quadrants and Breakdown




**Visuals and Graph Explanations:**

- **Velocity Scatter Plot:** Countries plotted across four quadrants using standardized performance scores and change rates. Emphasizes that high performance does not always coincide with high improvement.
- **Projected 2030 Gauge:** Shows projected SDG target achievement percentage. A tooltip clarifies that projections are based on linear trend extrapolation from historical data.
- **Country Performance Table (Goals 4 & 8):** Side-by-side comparison of educational attainment and youth employment indicators, allowing users to pinpoint cross-domain disparities.

## 6.1.5 Page 5 - Country and Demographic Breakdown




**Visuals and Graph Explanations**

- **SDG Indicator Progress Acceleration (Waterfall Chart):** Highlights year-by-year contributions to overall indicator progress.
- **Parity Gap Percentage by Country:** Bar chart of cumulative gender gaps, with tooltips clarifying that values aggregate multiple indicators and years.
- **Gender Parity Index (GPI) by Country:** Column chart representing relative female versus male performance across indicators, with values above 1 reflecting aggregated female advantage.

## 6.2 Methodological Enhancements

To ensure accuracy and interpretability, several refinements were applied during dashboard development:

1. **Indicator Aggregation Transparency:** Tooltips clarify that visualized values are aggregated or standardized across multiple indicators, countries, and years.
2. **Gender Parity Contextualization:** Explanatory tooltips for the Parity Gap and GPI charts prevent misinterpretation of values outside conventional ranges.
3. **Cross-SDG Comparison Methodology:** Performance scores are standardized within each SDG before aggregation, allowing valid comparative analysis across goals.
4. **Projection Disclosure:** The 2030 target gauge explicitly communicates that results are linear extrapolations of historical trends.
5. **Interactive Tooltip Use:** All methodological explanations are delivered via tooltips to preserve dashboard clarity while maintaining analytical transparency.

## 6.3 Outcome of Refinements

These enhancements collectively:

- Improve logical coherence and visual clarity.
- Support high-level comparative analyses of gender equity, educational outcomes, and country-level progress.
- Provide actionable and transparent insights for policymakers and other stakeholders.

# 7. Key Analytical Insights

## 7.1 Geographic Disparities and Top Performers

There were substantial regional disparities in SDG progress between 1990 and 2014, according to the study. Central Asian and Sub-Sahara African republics developed the most impressive upward development paths:

Azerbaijan: +67 (best progress globally)

Gambia: +60 points

Zimbabwe: +55 points

Botswana: +53 points

Uzbekistan: +51 points

These positive examples indicate that targeted policy actions, even in resource-constrained settings, can generate substantial gains in educational attainment outcomes and their contribution to youth employment. That the Northern region has a range (high to low) such that it may surpass the Southern region in absolute terms raises questions about traditional narratives of economic development and linear relationships for SDG progress.

## 7.2 Gender Equity Findings

- **Aggregate Trends:** On average at the international level, female quantitative indicators were marginally higher than male indicators (30.06 vs 24.17), with the gender gap reducing from approximately 10 points in 2000 to 7.5 points in 2023.
- **Critical Disparities:** Country-specific inequalities persist:
    - Pakistan: 500% parity gap (GPI ~6.2)
    - Bangladesh, India, Guatemala: Significant multi-dimensional gender disparities.
- **GPI Range:** Values differed from 0.2 to 6.0 and went beyond UNESCO equity corridor (0.97–1.03) indicating, in some countries, extreme disparities were present.

These findings emphasize both that the world has been making progress and that there are remaining uneven, regional gender gaps calling for specific policy interventions.

## 7.3 2030 Target Achievement Outlook

- **Projection Results:**
  - Current projected 2030 value: 18.30 (against aspirational target of 36.60)
  - Only 18.3% of countries are on track to meet SDG targets at current rates
- **Velocity Quadrant Analysis:** Most of the countries are centered on the "low performers, slow improvement," indicating that current speeds are infeasible for universal goal achievement by 2030.
- **Implications:** Linear extrapolations may have overestimated progress, without considering possible future disturbances, and the crusade therefore requires accelerated actions.

## 7.4 Temporal Volatility and External Shocks

Time series analysis shows large fluctuations because of the global events happened at that time:

- 2008: Global financial crisis affecting education funding and youth employment
- 2015: SDG framework adoption with measurement methodology changes
- 2020–2021: COVID-19 pandemic causing widespread educational disruption and youth unemployment

These factors justify the major fluctuation in the yearly analysis.

# 8. Challenges and Technical Solutions

## 8.1Data Quality and Outlier Management

- **Challenge:** Extreme outlier values for countries with low number of reports.
- **Solution:** Creating a DAX measure to allow large changes while restricting outliers.

## 8.2 Mixed Indicator Polarity

- **Challenge:** Difficulty in interpreting the 2 goals at the same time because in Goal 4, higher the better and in Goal 8, lower the better.
- **Solution:** Set the Goal 8 values to standard by normalizing by Performance_Score and then inverting change rates so that interpretations were consistent across all visuals.

## 8.3 Categorical Inconsistency and User Experience

- **Challenge:** Readability was obscured by complex technical codes (eg, LOWSECs, PRIMAR system status code, BOTHSEX)
- **Solution:** SQL-level normalization, Power Query calculated columns, and label visibility control for friendly end-user visualizations.

# 9. Reflective Analysis and Lessons Learned

## 9.1 Technical Architecture Insights

- **Data Modeling First Principle:** The Process of Normalization using SQL Server Management System and development of star schema design and DAX creation using Power BI.
- **Iterative Development Value:** Validation at each transformation stage to avoid schema errors and truncated text issues.
- **Quality Gates:** Filters such as Nature='C', Value IS NOT NULL to ensure reliable, country-level data integrity.

## 9.2 Analytical Methodology Considerations

- **Depth vs. Interpretability:** Tooltips and a clean layout to support advanced analyses and preserving clarity for non-specialists.
- **Trade-offs in aggregation:** Having standardized measurements that helps to avoid misinterpretation of heterogeneous indicators.
- **Project Limitations:** Linear trend in scenario is not predictive.

## 9.3 Stakeholder Communication and Dashboard Design

- **Tooltip based Transparency:** Based on the field and because of it, we had to find this balance between transparency and analytical efficiency that is the feature of our tooltips.
- **Value of Narrative Structure:** Multi-page structure allows for executive summaries
- **Color palette:** Warm monochromatic scheme adds professional look and accessibility.

## 9.4 Policy and Decision Support Implications

- Differentiated policy responses can target low performers, accelerate improvement in mid-level countries, and replicate best practices in high-performing nations.
- Aggregate measures of parity gaps reduce the reliance on judgements in interventions.
- Feasibility in future coverage can be estimated from projected achievement levels (e.g., a mere 18.3% are on target), which might lead to adjustments or enhanced interventions.

# 10. Conclusion

This development sets an example of SQL Server and Power BI working together as the foundation for tracking Sustainable Development Goals, from raw UN data to actionable intelligence.

**Technical Achievements:**

- Merged 27,000+ raw records into 23,487 facts with 100% referential integrity
- Star schema was optimized by 3-dimension tables.
- Created 20+ complex DAX measures in areas of time, cross-goal, predictive and equity analyses
- Have developed a 5-page dashboard with professional grade accessibility and customization features.

**Methodological Contributions:**

- Tooltip-based transparency ensures user understanding
- Aggregation problems were solved for better comparison and clarity
- Polarity conflicts solved for valid cross-goal comparison
- Projection assumptions were explicitly communicated

**Critical Findings:**

- Geographic variance: Azerbaijan (+67), Gambia (+60), Zimbabwe (+55) lead the way, other countries remain flat
- Gender gap remains despite global progress on closing it
- 2030 pace: 18.3% Only in five years is the requisite accelerated action not evident
- Temporal volatility: A focus on the resilience in SDG planning

**Policy Implications:**

- Serves the more cost-effective allocation of resources and stratified interventions
- Highlights best-practice replication opportunities
- Emphasizes need for systemic acceleration and statistical capacity investment

**Concluding Assessment**

Despite global progress, delivering on the transformational 2030 SDG agenda will require focused intervention toward well differentiated strategies and sustained investment in data infrastructure. This dashboard strikes a nimble balance between analytical sophistication, methodological transparency and stakeholder usability.